



Aortic Cross-Clamp & Cardioplegia

The heart is electrically silenced and protected — every minute of ischaemic time is counted

1 Closed loop communication

"Going for cross-clamp"

2 Cross-clamp applied

Surgeon applies aortic cross-clamp, isolating the heart from systemic flow. Ischaemic clock starts.

"Cross-clamp on"

3 Deliver cardioplegia

- Antegrade via aortic root and/or retrograde via coronary sinus
- Cold blood or crystalloid per unit protocol
- Monitor root/sinus pressure and delivered volume

CARDIOPLEGIA DOSING

Parameter	Target
Route	Antegrade / Retrograde
Temperature	Cold blood or crystalloid
Antegrade press.	100–150 mmHg
Retrograde press.	30–40 mmHg
Redose interval	Per unit protocol

4 Confirm arrest

- Electromechanical silence on ECG
- No mechanical activity on direct vision / TOE
- Myocardium cooled — topical ice/slush as needed

ARREST CONFIRMATION

- ☐ Flat ECG trace
- ☐ No mechanical activity
- ☐ Myocardium cooled / firm
- ☐ Cross-clamp time documented

5 Maintain protection

Repeat cardioplegia at protocol intervals. Communicate and document each dose, volume, and running cross-clamp time.

⚠ ISCHAEMIC TIME IS THE CLOCK THAT MATTERS

Every redose interval and the final total cross-clamp time must be communicated and documented.

6 Coming off cross-clamp

- De-air the heart
- Confirm rewarming underway

"Coming off cross-clamp"

7 Cross-clamp removed

Note total ischaemic time. Watch for return of cardiac activity; defibrillation may be required.

"Cross-clamp off"